A quantitative version of Chiti's inequality

Abstract

We prove a quantitative version of Chiti's reverse Hölder inequality. For every $p > q > 1$, its deficit controls, up to translations, the L^q-distance between the L^q-normalized first Dirichlet eigenfunction of a domain and that of the ball having the same first eigenvalue. In the range $1 < q \leq 2$ we also obtain a quantitative $W^{1,q}$ estimate. No upper bound on the measure of the domain is required.

1 Introduction

Symmetrization techniques provide a classical way of comparing solutions of partial differential equations on a general domain with radial solutions on a ball. In spectral problems, the Faber–Krahn inequality identifies balls as the minimizers of the first Dirichlet eigenvalue among sets with prescribed measure. A complementary point of view consists in fixing the first eigenvalue and comparing the concentration of the corresponding positive eigenfunctions. This is the setting of Chiti's reverse Hölder inequality [3, 4].

Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be a bounded, open and connected set. We denote by $\lambda = \lambda(\Omega)$ its principal Dirichlet eigenvalue, defined by

$$\lambda(\Omega) := \min_{v \in H_0^1(\Omega)\setminus\{0\}} \frac{\displaystyle\int_\Omega |\nabla v|^2 \, dx}{\displaystyle\int_\Omega v^2 \, dx}.$$

A minimizer $u \in H_0^1(\Omega)$ is called a principal Dirichlet eigenfunction of Ω. Equivalently, u is a nontrivial weak solution of

$$\begin{cases} -\Delta u = \lambda u & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The eigenfunction u can be chosen positive in Ω and is unique up to multiplication by a constant. We denote by B_λ the ball centred at the origin such that

$$\lambda(B_\lambda) = \lambda(\Omega) = \lambda,$$

and we let z be a positive principal Dirichlet eigenfunction of B_λ.

Chiti's inequality states that

$$\frac{\|u\|_{L^s(\Omega)}}{\|u\|_{L^r(\Omega)}} \leq \frac{\|z\|_{L^s(B_\lambda)}}{\|z\|_{L^r(B_\lambda)}} \qquad \text{for every } 0 < r \leq s \leq +\infty.$$

Thus, among domains having the same principal eigenvalue, the ball is the one whose principal eigenfunction is most concentrated. We fix $p > q > 1$ and normalize u and z by

$$\|u\|_{L^q(\Omega)} = \|z\|_{L^q(B_\lambda)} = 1.$$

For the pair (q, p), Chiti's inequality becomes

$$\|u\|^p_{L^p(\Omega)} \leq \|z\|^p_{L^p(B_\lambda)},$$

and we define the corresponding deficit by

$$\delta := \|z\|^p_{L^p(B_\lambda)} - \|u\|^p_{L^p(\Omega)}. \tag{1.1}$$

The purpose of this paper is to understand whether a small deficit forces the eigenfunction u to be close to the radial model z. There are two natural difficulties. First, the two functions are defined on different domains and the problem is invariant under translations. Secondly, fixing $\lambda(\Omega)$ does not provide an upper bound for $|\Omega|$: a domain may contain a large region on which its first eigenfunction is very small. The key observation is that the L^q normalization transfers the entire L^q-mass of this tail to the fixed interval corresponding to B_λ. This makes it possible to obtain a global L^q estimate without imposing any a priori bound on $|\Omega|$ and for every $q > 1$.

We denote by u^* and z^* the decreasing rearrangements of u and z, respectively, by $\Omega^\sharp$ the ball centred at the origin such that $\left|\Omega^\sharp\right| = |\Omega|$, and by $u^\sharp$ the Schwarz rearrangement of u. We set

$$\theta := \min\left\{\frac{1}{3}, \frac{1}{p+1}\right\}, \qquad b := |B_\lambda|,$$

and introduce the distinguished level and superlevel set

$$\bar{t} := u^*(b), \qquad U_{\bar{t}} := \{x \in \Omega : u(x) > \bar{t}\}.$$

By construction, $|U_{\bar{t}}| = |B_\lambda|$. More precisely, our main result can be stated as follows.

Theorem 1.1. *Let $n \geq 2$, let Ω be a bounded, open and connected set of $\mathbb{R}^n$, and let B_λ be the ball centred at the origin such that*

$$\lambda(B_\lambda) = \lambda(\Omega) =: \lambda.$$

Let $p > q > 1$, and let u and z be positive principal Dirichlet eigenfunctions of Ω and B_λ, respectively, normalized by

$$\|u\|_{L^q(\Omega)} = \|z\|_{L^q(B_\lambda)} = 1.$$

Set

$$\delta := \|z\|^p_{L^p(B_\lambda)} - \|u\|^p_{L^p(\Omega)}, \qquad \theta := \min\left\{\frac{1}{3}, \frac{1}{p+1}\right\}, \qquad \sigma := \theta \min\left\{1, \frac{q}{2}\right\}.$$

Then there exist two positive constants δ_0 and C, depending only on n, λ, p and q, such that, if $\delta \leq \delta_0$, there exists $x_0 \in \mathbb{R}^n$ for which

$$\|u - z(\cdot - x_0)\|^q_{L^q(\mathbb{R}^n)} \leq C\delta^\sigma. \tag{1.2}$$

Equivalently,

$$\|u - z(\cdot - x_0)\|_{L^q(\mathbb{R}^n)} \leq C\delta^{\frac{\theta}{\max\{2,q\}}}.$$

Here and in what follows, the eigenfunctions are extended by zero outside their domains.

In the subquadratic range the conclusion can be strengthened to the Sobolev norm. Notice that the exponent in the following statement is independent of q.

Theorem 1.2. *Under the assumptions of Theorem 1.1, suppose in addition that $1 < q \leq 2$. Then there exist two positive constants δ_0 and C, depending only on n, λ, p and q, such that, if $\delta \leq \delta_0$, there exists $x_0 \in \mathbb{R}^n$ for which*

$$\|u - z(\cdot - x_0)\|^q_{W^{1,q}(\mathbb{R}^n)} \leq C\delta^{q\theta/6}. \tag{1.3}$$

Equivalently,

$$\|u - z(\cdot - x_0)\|_{W^{1,q}(\mathbb{R}^n)} \leq C\delta^{\theta/6}.$$

The restriction $q \leq 2$ in Theorem 1.2 is structural for the present class of domains. For an arbitrary bounded open set a principal eigenfunction is a priori only in H^1_0, and no uniform $W^{1,q}$ regularity with $q > 2$ is available. In particular, the L^∞ interpolation used in the proof of Theorem 1.1 cannot be applied to the gradients without additional uniform regularity assumptions on the domains.

The constants in Theorem 1.1 are uniform among all bounded, open and connected sets having principal eigenvalue λ; in particular, they do not depend on $|\Omega|$. The two regimes in the exponent σ arise only when the L^2 eigenfunction stability for the Faber–Krahn inequality is converted into an L^q estimate on sets of fixed measure: Hölder's inequality is used for $q \leq 2$, while interpolation with a uniform L^∞ bound is used for $q \geq 2$.

As a direct consequence of Theorem 1.1 and of the estimate between the decreasing rearrangements, we also obtain that u is quantitatively close to a translate of its Schwarz rearrangement.

Corollary 1.3. *Under the assumptions of Theorem 1.1, there exist two positive constants δ_0 and C, depending only on n, λ, p and q, such that, if $\delta \leq \delta_0$, then*

$$\inf_{x_0 \in \mathbb{R}^n} \|u - u^\sharp(\,\cdot - x_0)\|^q_{L^q(\mathbb{R}^n)} \leq C\delta^\sigma. \tag{1.4}$$

Both functions are extended by zero outside their domains.

Let us briefly describe the proof. The first step is a weighted estimate for $z^* - u^*$, obtained by combining an elementary inequality between the powers p and q with the single-crossing property in Chiti's comparison theorem. We then exploit the L^q normalization to control the tail of u^* independently of $|\Omega|$ and deduce a global L^q estimate for the rearrangements. The boundary behaviour of z^* yields a bound for $\bar{t}$, and the truncation $(u - \bar{t})_+$ shows that $U_{\bar{t}}$ is almost optimal in the Faber–Krahn inequality. The quantitative L^2 eigenfunction stability of Allen, Kriventsov and Neumayer [1] compares the first eigenfunction of $U_{\bar{t}}$ with a translate of z. Since both comparison domains have fixed measure and their normalized eigenfunctions satisfy uniform L^∞ bounds, this estimate converts into an L^q estimate for every $q > 1$. Finally, a spectral-gap argument on $U_{\bar{t}}$ compares the truncation of u with that eigenfunction, while the global L^q tail estimate compares u with its truncation. For the Sobolev estimate we combine the Pólya–Szegő deficit with an integration-by-parts argument on B_λ. The gradient of the low-level tail is estimated without using $|\Omega|$ by testing the eigenvalue equation with a power of $\min\{u, \bar{t}\}$.

The paper is organized as follows. In Section 2 we recall the rearrangement notation and the comparison tools used throughout the paper. Section 3 contains the quantitative estimates for the Chiti deficit, the passage from the L^2 Faber–Krahn stability theorem to L^q, and the proof of the main theorem. Section 4 proves the $W^{1,q}$ estimate for $1 < q \leq 2$.

2 Preliminaries

For a non-negative measurable function f on a set of finite measure, we denote by

$$\mu_f(t) := |\{x : f(x) > t\}|$$

its distribution function and by f^* its decreasing rearrangement. The Schwarz rearrangement is the radial non-increasing function defined by

$$f^\sharp(x) := f^*\big(\omega_n |x|^n\big),$$

where ω_n denotes the measure of the unit ball of $\mathbb{R}^n$. Equimeasurability gives

$$\|f\|_{L^r} = \|f^*\|_{L^r} = \|f^\sharp\|_{L^r} \qquad \text{for every } r \in [1, +\infty].$$

By the Faber–Krahn inequality, $b = |B_\lambda| \le |\Omega|$. Accordingly, throughout the paper we extend z^* by zero on $(b, |\Omega|)$. Under the normalization

$$\|u\|_{L^q(\Omega)} = \|z\|_{L^q(B_\lambda)},$$

Chiti's comparison theorem gives the single-crossing property of u^* and z^* [3, 4]. In particular, the function

$$F(s) := \int_0^s \left((z^*)^q - (u^*)^q\right)\, dt$$

satisfies

$$F(s) \ge 0 \qquad \text{for every } s \in [0, |\Omega|], \qquad F(0) = F(|\Omega|) = 0.$$

We shall also use Chiti's L^∞ comparison,

$$\|u\|_{L^\infty(\Omega)} \le \|z\|_{L^\infty(B_\lambda)},$$

as well as domain monotonicity for the first Dirichlet eigenvalue and the Krahn–Szegő inequality for the second one. Finally, the quantitative Faber–Krahn inequality and its eigenfunction version will be used in the form recalled in Remark 3.7; see also [2, 1].

3 Estimates for the Chiti deficit

3.1 A weighted estimate for the rearrangements

Lemma 3.1. *Let $a, b \ge 0$ and $p > q > 1$. Then there exists a constant $c = c(p, q)$ such that*

$$a^p - b^p \ge \frac{p}{q} b^{p-q}(a^q - b^q) + c(a-b)^2(a+b)^{p-2}. \tag{3.1}$$

Proof. If $b = 0$ the inequality is trivially true with $c = 1$. Consider the case $b \ne 0$ and let $t = a/b$. Then

$$a^p - b^p - \frac{p}{q} b^{p-q}(a^q - b^q) = b^p \left[t^p - 1 - \frac{p}{q}(t^q - 1)\right],$$

and

$$(a-b)^2(a+b)^{p-2} = b^p (t-1)^2 (t+1)^{p-2}.$$

Thus it is sufficient to prove that the function $G : [0, +\infty) \to \mathbb{R}$ defined by

$$G(t) = \frac{t^p - 1 - \frac{p}{q}(t^q - 1)}{(t-1)^2(t+1)^{p-2}}$$

has a positive minimum. Indeed, the numerator is non-negative and vanishes only at $t = 1$. Moreover, both the numerator and its first derivative vanish at $t = 1$, whereas its second derivative at $t = 1$ is equal to $p(p-q) > 0$. Hence G admits a positive continuous extension at $t = 1$. Finally,

$$\lim_{t \to +\infty} G(t) = 1.$$

Therefore, by continuity, G has a positive minimum on $[0, +\infty)$. □

Proposition 3.2. *Let the assumptions of Theorem 1.1 hold. Denote by u^* and z^* the decreasing rearrangements of u and z, extend z^* by zero on $\big(|B_\lambda|, |\Omega|\big)$, and set*

$$\bar{t} := u^*\big(|B_\lambda|\big).$$

Then there exists a positive constant $c = c(n, \lambda, p, q)$ such that

$$\delta \geq c \int_0^{|\Omega|} (z^* - u^*)^2 (z^* + u^*)^{p-2}\, dt. \tag{3.2}$$

In particular, if $p \leq 2$, then

$$\delta \geq c \|z^* - u^*\|^2_{L^2(0,|\Omega|)}, \tag{3.3}$$

whereas, if $p > 2$, then

$$\delta \geq c\, \bar{t}^{p-2} \|z^* - u^*\|^2_{L^2(0,|B_\lambda|)}. \tag{3.4}$$

Proof. By the definition of δ and the equimeasurability of a function and its decreasing rearrangement, we have

$$\delta = \int_0^{|\Omega|} \left((z^*)^p - (u^*)^p\right)\, dt.$$

Therefore, by (3.1),

$$\begin{aligned} \delta &\geq c \int_0^{|\Omega|} (z^* - u^*)^2 (z^* + u^*)^{p-2}\, dt \\ &\quad + \frac{p}{q} \int_0^{|\Omega|} (u^*)^{p-q} \left((z^*)^q - (u^*)^q\right)\, dt. \end{aligned}$$

Set

$$F(s) := \int_0^s \left((z^*(t))^q - (u^*(t))^q\right)\, dt, \qquad s \in [0, |\Omega|].$$

The normalization of u and z gives

$$F(0) = F(|\Omega|) = 0,$$

and Chiti's comparison theorem yields

$$F(s) \geq 0 \qquad \text{for every } s \in [0, |\Omega|].$$

Moreover, the function $h := (u^*)^{p-q}$ is non-increasing. Hence, integrating by parts in the Riemann–Stieltjes sense, we obtain

$$\begin{aligned} \int_0^{|\Omega|} (u^*)^{p-q} F'\, dt &= \int_{[0,|\Omega|]} h\, dF \\ &= -\int_{[0,|\Omega|]} F\, dh \geq 0. \end{aligned}$$

This proves (3.2).

Suppose first that $p \leq 2$. By Chiti's L^∞ comparison,

$$u^*(t) \leq z^*(0) = \|z\|_{L^\infty(B_\lambda)} \qquad \text{for every } t \in [0, |\Omega|],$$

and clearly $z^*(t) \leq z^*(0)$. Since $p - 2 \leq 0$, it follows that

$$(z^* + u^*)^{p-2} \geq \left(2\|z\|_{L^\infty(B_\lambda)}\right)^{p-2}.$$

Estimate (3.3) now follows from (3.2).

Suppose now that $p > 2$. For every $t \in (0, |B_\lambda|)$, the monotonicity of u^* gives

$$u^*(t) \geq u^*\big(|B_\lambda|\big) = \bar{t}.$$

Consequently,

$$(z^*(t) + u^*(t))^{p-2} \geq \bar{t}^{p-2}.$$

Restricting the integral in (3.2) to $(0, |B_\lambda|)$ gives (3.4). □

Proposition 3.3. *Under the assumptions of Theorem 1.1, set*

$$b := |B_\lambda|, \qquad \bar{t} := u^*(b), \qquad \beta := \min\left\{\frac{1}{2}, \frac{q}{p}\right\}.$$

Then there exists a positive constant $C = C(n, \lambda, p, q)$ such that, if $\delta \le 1$, then

$$\int_b^{|\Omega|} (u^*)^q \, ds \le C\delta^\beta, \tag{3.5}$$

and

$$\|z^* - u^*\|^q_{L^q(0,|\Omega|)} \le C\delta^\beta. \tag{3.6}$$

Proof. Set

$$D := |z^* - u^*|, \qquad S := z^* + u^*, \qquad W := D^2 S^{p-2}.$$

By Chiti's L^∞ comparison, $S \le 2M$, where $M := \|z\|_{L^\infty(B_\lambda)}$. Moreover, the normalization of u and z and the fact that $z^* = 0$ in $(b, |\Omega|)$ give

$$\int_b^{|\Omega|} (u^*)^q \, ds = \int_0^b \left((z^*)^q - (u^*)^q\right) ds. \tag{3.7}$$

Since

$$|a^q - b^q| \le q|a - b|(a + b)^{q-1} \qquad \text{for every } a, b \ge 0,$$

we obtain

$$\int_b^{|\Omega|} (u^*)^q \, ds \le q \int_0^b DS^{q-1} \, ds. \tag{3.8}$$

Suppose first that $p \le 2q$. Then

$$DS^{q-1} = W^{1/2} S^{q-p/2}.$$

Since $q - p/2 \ge 0$, Hölder's inequality, the bound $S \le 2M$ and (3.2) yield

$$\int_0^b DS^{q-1} \, ds \le C \left(\int_0^b W \, ds\right)^{1/2} \le C\delta^{1/2}.$$

Suppose now that $p > 2q$. On the set $\{S > 0\}$, since $D \le S$, we have

$$DS^{q-1} = W^{q/p} \left(\frac{D}{S}\right)^{1-2q/p} \le W^{q/p}.$$

Thus, again by Hölder's inequality and (3.2),

$$\int_0^b DS^{q-1} \, ds \le b^{1-q/p} \left(\int_0^b W \, ds\right)^{q/p} \le C\delta^{q/p}.$$

Estimate (3.5) follows from (3.8).

We now estimate the L^q distance on $(0, b)$. If $p \le 2$, then $q < p \le 2$ and (3.3), together with Hölder's inequality, gives

$$\int_0^b D^q \, ds \le b^{1-q/2} \left(\int_0^b D^2 \, ds\right)^{q/2} \le C\delta^{q/2} \le C\delta^{1/2}.$$

If $p > 2$, then on $\{S > 0\}$,

$$D^q = W^{q/p} \left(\frac{D}{S}\right)^{q(1-2/p)} \leq W^{q/p},$$

and hence

$$\int_0^b D^q \, ds \leq b^{1-q/p} \left(\int_0^b W \, ds\right)^{q/p} \leq C\delta^{q/p}.$$

In both cases,

$$\int_0^b D^q \, ds \leq C\delta^{\beta}.$$

Since $z^* = 0$ on $(b, |\Omega|)$, adding (3.5) proves (3.6). □

3.2 The distinguished level and its superlevel set

Proposition 3.4. *Under the assumptions of Theorem 1.1, let $b = |B_\lambda|$ and $\bar{t} = u^*(b)$. Then there exists a positive constant $k = k(n, \lambda, q)$ such that*

$$\|z^* - u^*\|^2_{L^2(0,b)} \geq k\, \bar{t}^3. \tag{3.9}$$

Consequently, there exists a positive constant $C = C(n, \lambda, p, q)$ such that

$$\bar{t} \leq C\delta^{\theta}. \tag{3.10}$$

Proof. Since z is the positive principal eigenfunction of the ball, z^* is positive and decreasing in $[0, b)$ and satisfies

$$z^*(b) = 0.$$

Moreover, the regularity of z up to the boundary implies that

$$\sup_{s\in[0,b)} \frac{z^*(s)}{b - s} < +\infty.$$

Therefore, there exists a constant $K = K(n, \lambda, q) > 0$ such that

$$z^*(s) \leq K(b - s) \qquad \text{for every } s \in [0, b]. \tag{3.11}$$

Up to enlarging K, we may also assume that

$$K \geq \frac{1}{2b^{1+1/q}}.$$

Since u^* is non-increasing and $\|u\|_{L^q(\Omega)} = 1$, we have

$$b\,\bar{t}^q = b\big(u^*(b)\big)^q \leq \int_0^b (u^*)^q \, ds \leq 1.$$

It follows that

$$\bar{t} \leq b^{-1/q} \leq 2Kb.$$

We therefore define

$$h := \frac{\bar{t}}{2K},$$

which satisfies $0 < h \le b$. For every $s \in [b-h, b]$, the monotonicity of u^* gives $u^*(s) \ge \bar{t}$, whereas (3.11) gives $z^*(s) \le \bar{t}/2$. Consequently,

$$\begin{aligned}\|z^* - u^*\|^2_{L^2(0,b)} &\ge \int_{b-h}^{b} \big(u^*(s) - z^*(s)\big)^2 \, ds \\ &\ge \frac{\bar{t}^2}{4}\, h = \frac{1}{8K}\bar{t}^3,\end{aligned}$$

which proves (3.9) with $k = (8K)^{-1}$.

If $p \le 2$, combining (3.3) and (3.9) yields

$$\delta \ge c\|z^* - u^*\|^2_{L^2(0,b)} \ge ck\bar{t}^3.$$

If $p > 2$, combining (3.4) and (3.9) gives

$$\delta \ge c\bar{t}^{p-2}\|z^* - u^*\|^2_{L^2(0,b)} \ge ck\bar{t}^{p+1}.$$

These two estimates give (3.10). □

Lemma 3.5. *Under the assumptions of Theorem 1.1 there exist δ_1 and C, depending only on n, λ, p and q, such that, if $\delta \le \delta_1$, then*

$$\|u - (u-\bar{t})_+\|^q_{L^q(\Omega)} \le C\delta^\theta, \tag{3.12}$$

$$\|(u-\bar{t})_+\|_{L^q(U_{\bar{t}})} \ge \frac{1}{2}, \tag{3.13}$$

and

$$\|(u-\bar{t})_+\|_{L^2(U_{\bar{t}})} \ge c. \tag{3.14}$$

Proof. Set $v_{\bar{t}} := (u-\bar{t})_+$. By equimeasurability and the definition of $\bar{t}$,

$$\|u - v_{\bar{t}}\|^q_{L^q(\Omega)} = b\bar{t}^q + \int_b^{|\Omega|} (u^*)^q \, ds.$$

By (3.10) and (3.5),

$$\|u - v_{\bar{t}}\|^q_{L^q(\Omega)} \le C\delta^{q\theta} + C\delta^\beta \le C\delta^\theta.$$

Here we used $q > 1$, $\delta \le 1$, and $\beta \ge \theta$. The reverse triangle inequality now yields

$$\|v_{\bar{t}}\|_{L^q(U_{\bar{t}})} \ge 1 - C\delta^{\theta/q} \ge \frac{1}{2}$$

for δ sufficiently small.

It remains to obtain a lower bound in L^2. If $q \le 2$, Hölder's inequality on the set $U_{\bar{t}}$ of measure b gives

$$\|v_{\bar{t}}\|_{L^2(U_{\bar{t}})} \ge b^{1/2-1/q}\|v_{\bar{t}}\|_{L^q(U_{\bar{t}})} \ge c.$$

If $q > 2$, Chiti's L^∞ comparison gives $v_{\bar{t}} \le u \le M := \|z\|_{L^\infty(B_\lambda)}$, and therefore

$$\|v_{\bar{t}}\|^2_{L^2(U_{\bar{t}})} \ge M^{2-q}\|v_{\bar{t}}\|^q_{L^q(U_{\bar{t}})} \ge c.$$

This proves all the assertions. □

Corollary 3.6. *Let the assumptions and the notation of Theorem 1.1 hold. Then there exist constants $\delta_2 > 0$ and $C > 0$, depending only on the parameters in Theorem 1.1, such that, if $\delta \le \delta_2$, then*

$$0 \le \lambda(U_{\bar{t}}) - \lambda(\Omega) \le C\bar{t} \le C\delta^\theta. \tag{3.15}$$

Proof. Set $v_{\bar t} := (u - \bar t)_+$. Since $v_{\bar t} \in H^1_0(U_{\bar t})$, it is an admissible competitor in the Rayleigh characterization of $\lambda(U_{\bar t})$. Moreover, testing the eigenvalue equation for u with $v_{\bar t}$ gives

$$\begin{aligned}\int_{U_{\bar t}} |\nabla v_{\bar t}|^2 \, dx &= \lambda \int_{U_{\bar t}} u v_{\bar t} \, dx \\ &= \lambda \|v_{\bar t}\|^2_{L^2(U_{\bar t})} + \lambda \bar t \int_{U_{\bar t}} v_{\bar t} \, dx.\end{aligned}$$

Consequently,

$$\lambda(U_{\bar t}) - \lambda \le \lambda \bar t \, \frac{\displaystyle\int_{U_{\bar t}} v_{\bar t} \, dx}{\|v_{\bar t}\|^2_{L^2(U_{\bar t})}}.$$

By Hölder's inequality,

$$\int_{U_{\bar t}} v_{\bar t} \, dx \le b^{1/2} \|v_{\bar t}\|_{L^2(U_{\bar t})}.$$

On the other hand, Lemma 3.5 gives, after reducing δ_2 if necessary,

$$\|v_{\bar t}\|^2_{L^2(U_{\bar t})} \ge c.$$

It follows that

$$0 \le \lambda(U_{\bar t}) - \lambda(\Omega) \le C \bar t.$$

The last inequality in (3.15) follows from (3.10). □

3.3 Comparison with the model eigenfunction

Remark 3.7. The quantitative eigenfunction stability for the Faber–Krahn inequality proved in [1, Theorem 1.1] applies to $U_{\bar t}$. Indeed,

$$\lambda(\Omega) = \lambda(B_\lambda) \qquad \text{and} \qquad |U_{\bar t}| = |B_\lambda|.$$

Let $w_{\bar t}$ be the positive principal Dirichlet eigenfunction of $U_{\bar t}$, normalized so that

$$\|w_{\bar t}\|_{L^q(U_{\bar t})} = \|z\|_{L^q(B_\lambda)} = 1.$$

Then, extending both eigenfunctions by zero outside their domains, there exist $x_{\bar t} \in \mathbb{R}^n$ and $c = c(n, \lambda) > 0$ such that

$$\lambda(U_{\bar t}) - \lambda(B_\lambda) \ge c \, \Big\| \frac{w_{\bar t}}{\|w_{\bar t}\|_{L^2(U_{\bar t})}} - \frac{z(\,\cdot - x_{\bar t})}{\|z\|_{L^2(B_\lambda)}} \Big\|^2_{L^2(\mathbb{R}^n)}. \tag{3.16}$$

Corollary 3.8. *Under the assumptions of Theorem 1.1, provided δ is sufficiently small, there exists $x_{\bar t} \in \mathbb{R}^n$ such that*

$$\|w_{\bar t} - z(\,\cdot - x_{\bar t})\|^q_{L^q(\mathbb{R}^n)} \le C \delta^\sigma, \qquad \sigma := \theta \min\Big\{1, \frac{q}{2}\Big\}. \tag{3.17}$$

Proof. Set

$$\widehat{w}_{\bar t} := \frac{w_{\bar t}}{\|w_{\bar t}\|_{L^2(U_{\bar t})}}, \qquad \widehat{z} := \frac{z}{\|z\|_{L^2(B_\lambda)}}.$$

Combining (3.16) with (3.15), we obtain

$$\|\widehat{w}_{\bar t} - \widehat{z}(\,\cdot - x_{\bar t})\|^2_{L^2(\mathbb{R}^n)} \le C \delta^\theta. \tag{3.18}$$

Both functions are supported on sets of measure b. Moreover, the standard L^2–L^∞ estimate for Dirichlet eigenfunctions gives

$$\|\widehat{w}_{\bar{t}}\|_{L^\infty(U_{\bar{t}})} \le C_n \lambda(U_{\bar{t}})^{n/4} \le C, \qquad \|\widehat{z}\|_{L^\infty(B_\lambda)} \le C.$$

If $1 < q \le 2$, Hölder's inequality on the union of the two supports yields

$$\|\widehat{w}_{\bar{t}} - \widehat{z}(\,\cdot - x_{\bar{t}})\|_{L^q}^q \le (2b)^{1-q/2} \|\widehat{w}_{\bar{t}} - \widehat{z}(\,\cdot - x_{\bar{t}})\|_{L^2}^q \le C\delta^{q\theta/2}.$$

If $q \ge 2$, interpolation with the uniform L^∞ bound gives

$$\|\widehat{w}_{\bar{t}} - \widehat{z}(\,\cdot - x_{\bar{t}})\|_{L^q}^q \le C \|\widehat{w}_{\bar{t}} - \widehat{z}(\,\cdot - x_{\bar{t}})\|_{L^2}^2 \le C\delta^{\theta}.$$

Thus, in both cases,

$$\|\widehat{w}_{\bar{t}} - \widehat{z}(\,\cdot - x_{\bar{t}})\|_{L^q}^q \le C\delta^{\sigma}. \tag{3.19}$$

The L^q norms of $\widehat{w}_{\bar{t}}$ and $\widehat{z}$ are bounded from below by a positive constant depending only on n, λ and q. For $q \ge 2$ this follows from Hölder's inequality and the fixed measure of the supports; for $1 < q < 2$ it follows from their L^2 normalization and the uniform L^∞ bound. Applying the elementary normalization estimate

$$\Big\| \frac{f}{\|f\|_{L^q}} - \frac{g}{\|g\|_{L^q}} \Big\|_{L^q} \le \frac{2}{\|f\|_{L^q}} \|f - g\|_{L^q}$$

and recalling that $w_{\bar{t}}$ and z are L^q-normalized, (3.17) follows from (3.19). □

Proposition 3.9. *Under the assumptions of Theorem 1.1, let $w_{\bar{t}}$ be a positive principal Dirichlet eigenfunction of $U_{\bar{t}}$, normalized by*

$$\|w_{\bar{t}}\|_{L^q(U_{\bar{t}})} = 1,$$

and extended by zero outside $U_{\bar{t}}$. Then there exist two positive constants δ_3 and C, depending only on n, λ, p and q, such that, if $\delta \le \delta_3$, then

$$\|u - w_{\bar{t}}\|_{L^q(\Omega)}^q \le C\delta^{\sigma}. \tag{3.20}$$

Proof. Set

$$v_{\bar{t}} := (u - \bar{t})_+, \qquad \widehat{w}_{\bar{t}} := \frac{w_{\bar{t}}}{\|w_{\bar{t}}\|_{L^2(U_{\bar{t}})}}.$$

We project $v_{\bar{t}}$ onto the first eigenspace of $U_{\bar{t}}$ by defining

$$a_{\bar{t}} := \int_{U_{\bar{t}}} v_{\bar{t}} \widehat{w}_{\bar{t}}\, dx, \qquad r_{\bar{t}} := v_{\bar{t}} - a_{\bar{t}} \widehat{w}_{\bar{t}}.$$

Since $v_{\bar{t}}$ and $\widehat{w}_{\bar{t}}$ are nonnegative and nontrivial, $a_{\bar{t}} > 0$, and

$$\int_{U_{\bar{t}}} r_{\bar{t}} \widehat{w}_{\bar{t}}\, dx = 0.$$

Thus $r_{\bar{t}}$ is admissible in the variational characterization of $\lambda_2(U_{\bar{t}})$, and hence

$$\begin{aligned} &\big(\lambda_2(U_{\bar{t}}) - \lambda_1(U_{\bar{t}})\big) \|r_{\bar{t}}\|_{L^2(U_{\bar{t}})}^2 \\ &\qquad \le \int_{U_{\bar{t}}} |\nabla r_{\bar{t}}|^2\, dx - \lambda_1(U_{\bar{t}}) \|r_{\bar{t}}\|_{L^2(U_{\bar{t}})}^2. \end{aligned} \tag{3.21}$$

Since $\widehat{w}_{\bar{t}}$ is an L^2-normalized first eigenfunction and $r_{\bar{t}}$ is obtained by orthogonal projection, we have

$$\begin{aligned}\int_{U_{\bar{t}}} |\nabla r_{\bar{t}}|^2 \, dx - \lambda_1(U_{\bar{t}}) \|r_{\bar{t}}\|^2_{L^2(U_{\bar{t}})} \\ = \int_{U_{\bar{t}}} |\nabla v_{\bar{t}}|^2 \, dx - \lambda_1(U_{\bar{t}}) \|v_{\bar{t}}\|^2_{L^2(U_{\bar{t}})}.\end{aligned} \tag{3.22}$$

By the Krahn–Szegő inequality and the identity $|U_{\bar{t}}| = |B_\lambda|$,

$$\lambda_2(U_{\bar{t}}) \geq 2^{2/n} \lambda.$$

On the other hand, (3.15) gives

$$\lambda_1(U_{\bar{t}}) \leq \lambda + C\delta^\theta.$$

Therefore, after reducing δ_3 if necessary,

$$\lambda_2(U_{\bar{t}}) - \lambda_1(U_{\bar{t}}) \geq \frac{1}{2}\left(2^{2/n} - 1\right)\lambda =: g > 0. \tag{3.23}$$

Testing the eigenvalue equation for u with $v_{\bar{t}}$ gives

$$\int_{U_{\bar{t}}} |\nabla v_{\bar{t}}|^2 \, dx = \lambda \|v_{\bar{t}}\|^2_{L^2(U_{\bar{t}})} + \lambda \bar{t} \int_{U_{\bar{t}}} v_{\bar{t}} \, dx.$$

Since $\lambda_1(U_{\bar{t}}) \geq \lambda$ by domain monotonicity and

$$\int_{U_{\bar{t}}} v_{\bar{t}} \, dx \leq b^{1-1/q} \|v_{\bar{t}}\|_{L^q(U_{\bar{t}})} \leq b^{1-1/q},$$

we obtain

$$0 \leq \int_{U_{\bar{t}}} |\nabla v_{\bar{t}}|^2 \, dx - \lambda_1(U_{\bar{t}}) \|v_{\bar{t}}\|^2_{L^2(U_{\bar{t}})} \leq C\bar{t} \leq C\delta^\theta. \tag{3.24}$$

Combining (3.21), (3.22), (3.23) and (3.24), we find

$$\|v_{\bar{t}} - a_{\bar{t}} \widehat{w}_{\bar{t}}\|^2_{L^2(U_{\bar{t}})} \leq C\delta^\theta. \tag{3.25}$$

By Lemma 3.5, $\|v_{\bar{t}}\|_{L^2(U_{\bar{t}})} \geq c$. Applying the elementary normalization estimate to $v_{\bar{t}}$ and $a_{\bar{t}}\widehat{w}_{\bar{t}}$, and recalling that $a_{\bar{t}} > 0$, we deduce

$$\left\| \frac{v_{\bar{t}}}{\|v_{\bar{t}}\|_{L^2(U_{\bar{t}})}} - \widehat{w}_{\bar{t}} \right\|^2_{L^2(U_{\bar{t}})} \leq C\delta^\theta. \tag{3.26}$$

Set

$$\widehat{v}_{\bar{t}} := \frac{v_{\bar{t}}}{\|v_{\bar{t}}\|_{L^2(U_{\bar{t}})}}.$$

The functions $\widehat{v}_{\bar{t}}$ and $\widehat{w}_{\bar{t}}$ are supported in $U_{\bar{t}}$, which has measure b, and are uniformly bounded in L^∞. Indeed, the L^∞ bound for $\widehat{v}_{\bar{t}}$ follows from Chiti's comparison and the lower bound for $\|v_{\bar{t}}\|_{L^2}$, while the bound for $\widehat{w}_{\bar{t}}$ follows from the standard eigenfunction estimate and the uniform upper bound for $\lambda(U_{\bar{t}})$. Thus, arguing exactly as in the proof of Corollary 3.8, estimate (3.26) implies

$$\|\widehat{v}_{\bar{t}} - \widehat{w}_{\bar{t}}\|^q_{L^q(U_{\bar{t}})} \leq C\delta^\sigma. \tag{3.27}$$

The L^q norms of these two L^2-normalized functions are bounded from below uniformly. Applying the L^q normalization estimate to (3.27), and recalling that $\|w_{\bar{t}}\|_{L^q} = 1$, gives

$$\left\| \frac{v_{\bar{t}}}{\|v_{\bar{t}}\|_{L^q(U_{\bar{t}})}} - w_{\bar{t}} \right\|^q_{L^q(U_{\bar{t}})} \leq C\delta^\sigma. \tag{3.28}$$

It remains to remove the truncation. By (3.12),

$$\|u - v_{\bar{t}}\|_{L^q(\Omega)}^q \le C\delta^\theta.$$

Since $\|u\|_{L^q} = 1$, the reverse triangle inequality yields

$$|1 - \|v_{\bar{t}}\|_{L^q}| \le \|u - v_{\bar{t}}\|_{L^q}.$$

Consequently,

$$\|u - \frac{v_{\bar{t}}}{\|v_{\bar{t}}\|_{L^q(U_{\bar{t}})}}\|_{L^q(\Omega)}^q \le C\delta^\theta \le C\delta^\sigma. \tag{3.29}$$

Finally, (3.20) follows from (3.28), (3.29), and the triangle inequality. □

Proof of Theorem 1.1. By Corollary 3.8, there exists $x_0 \in \mathbb{R}^n$ such that

$$\|w_{\bar{t}} - z(\cdot - x_0)\|_{L^q(\mathbb{R}^n)}^q \le C\delta^\sigma.$$

On the other hand, Proposition 3.9 gives

$$\|u - w_{\bar{t}}\|_{L^q(\mathbb{R}^n)}^q \le C\delta^\sigma.$$

Therefore, by the triangle inequality,

$$\|u - z(\cdot - x_0)\|_{L^q(\mathbb{R}^n)}^q \le 2^{q-1}\left(\|u - w_{\bar{t}}\|_{L^q}^q + \|w_{\bar{t}} - z(\cdot - x_0)\|_{L^q}^q\right) \le C\delta^\sigma.$$

This proves (1.2). □

Proof of Corollary 1.3. By (3.6),

$$\|u^\sharp - z\|_{L^q(\mathbb{R}^n)}^q = \|u^* - z^*\|_{L^q(0,|\Omega|)}^q \le C\delta^\beta, \qquad \beta := \min\left\{\frac{1}{2}, \frac{q}{p}\right\}.$$

Since $\beta \ge \theta \ge \sigma$ and $\delta \le 1$, the right-hand side is bounded by $C\delta^\sigma$. Let $x_0 \in \mathbb{R}^n$ be the point provided by Theorem 1.1. By translation invariance, the preceding estimate gives

$$\|u^\sharp(\cdot - x_0) - z(\cdot - x_0)\|_{L^q(\mathbb{R}^n)}^q \le C\delta^\sigma.$$

Combining this estimate with (1.2) and using the triangle inequality, we obtain

$$\|u - u^\sharp(\cdot - x_0)\|_{L^q(\mathbb{R}^n)}^q \le C\delta^\sigma.$$

This proves (1.4). □

4 Sobolev stability in the range $1 < q \le 2$

Throughout this section we assume $1 < q \le 2$ and use the notation of Theorem 1.1. We also set

$$\beta := \min\left\{\frac{1}{2}, \frac{q}{p}\right\}.$$

Recall that $\beta \ge \theta$.

We first recover the L^2-normalized estimates which are needed in the gradient argument. This step is independent of the conversion from L^2 to L^q used in the proof of Theorem 1.1.

Lemma 4.1. *Let*

$$\widehat{u} := \frac{u}{\|u\|_{L^2(\Omega)}}, \qquad \widehat{z} := \frac{z}{\|z\|_{L^2(B_\lambda)}}, \qquad \widehat{u^\sharp} := \frac{u^\sharp}{\|u^\sharp\|_{L^2(\Omega^\sharp)}}.$$

There exists $x_0 \in \mathbb{R}^n$ such that

$$\begin{aligned} \|\widehat{u} - \widehat{z}(\,\cdot - x_0)\|^2_{L^2(\mathbb{R}^n)} + \|\widehat{u^\sharp} - \widehat{z}\|^2_{L^2(\mathbb{R}^n)} \\ + \|\widehat{u} - \widehat{u^\sharp}(\,\cdot - x_0)\|^2_{L^2(\mathbb{R}^n)} \le C\delta^\theta. \end{aligned} \tag{4.1}$$

Moreover, the L^2 norms of u, $u^\sharp$ and z are bounded from above and below by positive constants depending only on the fixed parameters.

Proof. The upper bounds follow from Chiti's L^∞ comparison and the L^q normalization. The lower bound for u follows from (3.14), while the corresponding statements for $u^\sharp$ and z are immediate.

Let $v_{\bar t} = (u - \bar t)_+$. Since $u \le \|z\|_{L^\infty(B_\lambda)}$ and $u^* \le \bar t$ on $(b, |\Omega|)$, estimates (3.5) and (3.10) give

$$\begin{aligned} \|u - v_{\bar t}\|^2_{L^2(\Omega)} &= b\bar t^2 + \int_b^{|\Omega|} (u^*)^2 \, ds \\ &\le b\bar t^2 + \bar t^{2-q} \int_b^{|\Omega|} (u^*)^q \, ds \le C\delta^\theta. \end{aligned} \tag{4.2}$$

Here and below the case $q = 2$ is understood with $\bar t^{2-q} = 1$. The normalization estimate, (3.26), and (3.18) therefore imply, for the point x_0 supplied by the quantitative Faber–Krahn estimate,

$$\|\widehat{u} - \widehat{z}(\,\cdot - x_0)\|^2_{L^2(\mathbb{R}^n)} \le C\delta^\theta.$$

On the other hand, $\big|u^\sharp - z\big| \le 2\|z\|_{L^\infty(B_\lambda)}$ and (3.6) yield

$$\|u^\sharp - z\|^2_{L^2(\mathbb{R}^n)} \le \big(2\|z\|_{L^\infty(B_\lambda)}\big)^{2-q} \, \|u^\sharp - z\|^q_{L^q(\mathbb{R}^n)} \le C\delta^\beta \le C\delta^\theta.$$

After normalization this gives the second estimate in (4.1). The third one follows from the first two estimates and the triangle inequality. □

The next estimate is the quantitative Pólya–Szegő deficit used in the old H^1 argument.

Lemma 4.2. *Under the assumptions of Theorem 1.2,*

$$0 \le \int_\Omega |\nabla u|^2 \, dx - \int_{\Omega^\sharp} \left|\nabla u^\sharp\right|^2 dx \le C\delta^\theta. \tag{4.3}$$

The same estimate holds for the L^2-normalized functions.

Proof. Set $v := (u^\sharp - \bar t)_+$. Since $\big|\{u^\sharp > \bar t\}\big| = b$, the support of v is B_λ and $v \in H^1_0(B_\lambda)$. Hence, by the variational characterization of $\lambda(B_\lambda)$,

$$\int_{B_\lambda} |\nabla v|^2 \, dx \ge \lambda \int_{B_\lambda} v^2 \, dx.$$

Moreover, $|\nabla v| = \left|\nabla u^\sharp\right|$ a.e. in $\{u^\sharp > \bar{t}\}$ and vanishes elsewhere. Therefore, by the Pólya–Szegő inequality,

$$\begin{aligned}
0 &\le \int_\Omega |\nabla u|^2\,dx - \int_{\Omega^\sharp} \left|\nabla u^\sharp\right|^2 dx \\
&\le \int_\Omega |\nabla u|^2\,dx - \int_{B_\lambda} |\nabla v|^2\,dx \\
&\le \lambda\left(\int_\Omega u^2\,dx - \int_{B_\lambda} v^2\,dx\right) \\
&= \lambda\left[2\bar{t}\int_0^b u^*(s)\,ds - b\bar{t}^2 + \int_b^{|\Omega|} (u^*(s))^2\,ds\right].
\end{aligned}$$

The first term is bounded by $C\bar{t}$, while the last one is bounded as in (4.2). This proves (4.3). Since u and $u^\sharp$ have the same L^2 norm and this norm is bounded from below, division by that norm proves the normalized version. □

We now correct and record the integration-by-parts part of the H^1 argument. In particular, the quantities on the left-hand side below are squared gradient norms.

Lemma 4.3. *For the point x_0 in Lemma 4.1,*

$$\|\nabla u - \nabla u^\sharp(\,\cdot - x_0)\|^2_{L^2(\mathbb{R}^n)} \le C\delta^{\theta/3}, \tag{4.4}$$

and

$$\|\nabla u^\sharp - \nabla z\|^2_{L^2(\mathbb{R}^n)} \le C\delta^{\theta}. \tag{4.5}$$

Proof. Write

$$a := \|u\|_{L^2(\Omega)} = \|u^\sharp\|_{L^2(\Omega^\sharp)}, \qquad c := \|z\|_{L^2(B_\lambda)}.$$

By Lemma 4.1, both numbers are bounded above and below. Let

$$U := \frac{u^\sharp}{a}, \qquad Z := \frac{z}{c}.$$

Both functions have unit L^2 norm. Since Z is radial, its outer normal derivative on ∂B_λ is a negative constant, denoted by Z_N. The Pólya–Szegő inequality and integration by parts on B_λ give

$$\begin{aligned}
\|\nabla U - \nabla Z\|^2_{L^2(\mathbb{R}^n)} &\le 2\lambda - 2\int_{B_\lambda} \nabla U\cdot\nabla Z\,dx \\
&= 2\lambda - 2\lambda\int_{B_\lambda} UZ\,dx - 2Z_N\int_{\partial B_\lambda} U\,d\mathcal{H}^{n-1} \\
&= \lambda\|U - Z\|^2_{L^2(\mathbb{R}^n)} - 2Z_N|\partial B_\lambda|\frac{\bar{t}}{a} \le C\delta^\theta.
\end{aligned}$$

In the last line we used (4.1) and (3.10). Furthermore,

$$|a - c| \le \|u^\sharp - z\|_{L^2(\mathbb{R}^n)} \le C\delta^{\beta/2}.$$

Consequently,

$$\|\nabla u^\sharp - \nabla z\|_{L^2(\mathbb{R}^n)} \le a\|\nabla U - \nabla Z\|_{L^2(\mathbb{R}^n)} + |a - c|\|\nabla Z\|_{L^2(\mathbb{R}^n)},$$

which proves (4.5).

We now prove (4.4). Set

$$F := \frac{u}{a}, \qquad G := \frac{u^\sharp(\cdot - x_0)}{a}, \qquad Z_0 := Z(\cdot - x_0), \qquad h := F - G,$$

and let $B_0 := B_\lambda(x_0)$. Define

$$H := \|h\|_{L^2(\mathbb{R}^n)}, \qquad X := \|\nabla h\|^2_{L^2(\mathbb{R}^n)}.$$

By (4.1), $H \le C\delta^{\theta/2}$. Moreover, the normalized version of (4.3), together with the identity

$$X = \|\nabla F\|^2_{L^2} - \|\nabla G\|^2_{L^2} - 2\int_{\mathbb{R}^n} \nabla G \cdot \nabla h \, dx,$$

gives

$$X \le C\delta^\theta + 2\left|\int_{\mathbb{R}^n} \nabla(G - Z_0) \cdot \nabla h \, dx\right| + 2\left|\int_{B_0} \nabla Z_0 \cdot \nabla h \, dx\right|. \tag{4.6}$$

By (4.5) and the uniform bounds on a, the first cross term is bounded, by Young's inequality, by

$$C\delta^\theta + \frac{1}{8}X.$$

For the second one, integration by parts on B_0 gives

$$\int_{B_0} \nabla Z_0 \cdot \nabla h \, dx = \lambda \int_{B_0} Z_0 h \, dx + Z_N \int_{\partial B_0} h \, d\mathcal{H}^{n-1}.$$

The interior term is bounded by CH. The interpolation trace inequality on the fixed ball B_0 yields

$$\|h\|_{L^2(\partial B_0)} \le C\|h\|^{1/2}_{L^2(B_0)}\|h\|^{1/2}_{H^1(B_0)} \le CH^{1/2}(H^2 + X)^{1/4}.$$

Using Young's inequality,

$$H^{1/2}(H^2 + X)^{1/4} \le CH + CH^{1/2}X^{1/4} \le CH + CH^{2/3} + \frac{1}{8}X.$$

Combining these estimates with (4.6) and using $H \le C\delta^{\theta/2}$, we obtain

$$X \le C\delta^{\theta/3}.$$

Finally, multiplying by the uniformly bounded factor a proves (4.4). □

To pass from H^1 to $W^{1,q}$ without introducing a factor depending on $|\Omega|$, we need a separate estimate on the low-level tail.

Lemma 4.4. *Set*

$$\eta := \min\{u, \bar{t}\}, \qquad \eta^\sharp := \min\{u^\sharp, \bar{t}\}.$$

Then

$$\|\nabla\eta\|^2_{L^2(\mathbb{R}^n)} + \|\nabla\eta^\sharp\|^2_{L^2(\mathbb{R}^n)} \le C\delta^\theta \tag{4.7}$$

and

$$\|\nabla\eta\|^q_{L^q(\mathbb{R}^n)} + \|\nabla\eta^\sharp\|^q_{L^q(\mathbb{R}^n)} \le C\delta^{q\theta/6}. \tag{4.8}$$

Proof. Testing $-\Delta u = \lambda u$ with η gives

$$\|\nabla \eta\|^2_{L^2(\mathbb{R}^n)} = \lambda \left(\int_{\{u<\bar{t}\}} u^2 \, dx + \bar{t} \int_{\{u\geq \bar{t}\}} u \, dx \right).$$

The first term is at most

$$\bar{t}^{2-q} \int_b^{|\Omega|} (u^*)^q \, ds,$$

while the second one is bounded by $C\bar{t}$, because $\big|\{u \geq \bar{t}\}\big| = b$. Thus (3.5) and (3.10) give

$$\|\nabla \eta\|^2_{L^2(\mathbb{R}^n)} \leq C\delta^\theta.$$

Since rearrangement commutes with truncation, the Pólya–Szegő inequality gives the same estimate for $\eta^\sharp$, proving (4.7).

If $q = 2$, (4.8) follows directly from (4.7). Assume now $1 < q < 2$. Testing the eigenvalue equation with $\min\{u, \bar{t}\}^{q-1}$ gives

$$(q-1) \int_{\{u<\bar{t}\}} u^{q-2} |\nabla u|^2 \, dx = \lambda \left(\int_{\{u<\bar{t}\}} u^q \, dx + \bar{t}^{q-1} \int_{\{u\geq \bar{t}\}} u \, dx \right) \leq C\delta^{(q-1)\theta}. \tag{4.9}$$

Here we used $\beta \geq \theta > (q-1)\theta$. By Hölder's inequality,

$$\begin{aligned} \int_{\{u<\bar{t}\}} |\nabla u|^q \, dx &\leq \left(\int_{\{u<\bar{t}\}} u^{q-2} |\nabla u|^2 \, dx \right)^{q/2} \left(\int_{\{u<\bar{t}\}} u^q \, dx \right)^{(2-q)/2} \\ &\leq C\delta^{\frac{q(q-1)}{2}\theta + \frac{2-q}{2}\beta} \leq C\delta^{q\theta/6}. \end{aligned}$$

For the last inequality we used $\beta \geq \theta$ and

$$\frac{1}{2}(q^2 - 2q + 2) \geq \frac{q}{6}, \qquad 1 < q < 2.$$

Finally, set

$$\Phi(s) := \frac{2}{q} \min\{s, \bar{t}\}^{q/2}.$$

Since $\Phi(u)^\sharp = \Phi(u^\sharp)$, the Pólya–Szegő inequality applied to $\Phi(u)$ shows that the weighted integral on the left-hand side of (4.9) does not increase under rearrangement. The same Hölder argument therefore proves the estimate for $\eta^\sharp$. □

Proof of Theorem 1.2. Let x_0 be the point in Lemma 4.3, and set

$$v := (u - \bar{t})_+, \qquad v^\sharp := (u^\sharp - \bar{t})_+.$$

The supports of v and $v^\sharp(\,\cdot - x_0)$ have measure b. By (4.4) and (4.7),

$$\|\nabla v - \nabla v^\sharp(\,\cdot - x_0)\|_{L^2(\mathbb{R}^n)} \leq C\delta^{\theta/6}.$$

Hölder's inequality on the union of the two supports, whose measure is at most $2b$, yields

$$\|\nabla v - \nabla v^\sharp(\,\cdot - x_0)\|^q_{L^q(\mathbb{R}^n)} \leq C\delta^{q\theta/6}.$$

Adding back the two tails and using (4.8), we obtain

$$\|\nabla u - \nabla u^\sharp(\,\cdot - x_0)\|^q_{L^q(\mathbb{R}^n)} \leq C\delta^{q\theta/6}. \tag{4.10}$$

Similarly, $v^\sharp$ and z are supported in B_λ. Estimates (4.5) and (4.7) give

$$\|\nabla v^\sharp - \nabla z\|_{L^2(\mathbb{R}^n)} \leq C\delta^{\theta/2}.$$

Using Hölder's inequality on B_λ and then adding back $\eta^\sharp$, we find

$$\|\nabla u^\sharp - \nabla z\|^q_{L^q(\mathbb{R}^n)} \leq C\delta^{q\theta/6}. \tag{4.11}$$

The zero-order terms are better: (1.4) and (3.6), together with $\beta \geq \theta$, imply

$$\|u - u^\sharp(\cdot - x_0)\|^q_{L^q} + \|u^\sharp - z\|^q_{L^q} \leq C\delta^{q\theta/6}.$$

Combining this estimate with (4.10) and (4.11), and translating the latter by x_0, proves (1.3). □

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